Interplay of galaxy formation and the evolution of dark matter haloes in the cosmic web

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IUCAA

List of Publications

1. Velmani & Paranjape, 2023, "The quasi-adiabatic relaxation of haloes in the IllustrisTNG and EAGLE cosmological simulations", **MNRAS**, 520(2):2867-2886. doi:10.1093/mnras/stad297

2. Velmani & Paranjape, 2024, "A self-similar model of galaxy formation and dark halo relaxation", **JCAP**, doi:10.1088/1475-7516/2024/05/080,

3. Velmani & Paranjape, 2025,"Dynamics of the response of dark matter halo to galaxy evolution in IllustrisTNG", **JCAP**, doi:10.1088/1475-7516/2025/02/006,

4. Velmani & Paranjape, 2024,"Role of astrophysical modeling on dark matter halo relaxation response at redshifts $z = 0$ and $z = 1$", arXiv:2408.04864, **JCAP** *subm.*

Not included in this thesis

1. Ramakrishnan & Velmani, 2022, "Properties beyond mass for unresolved haloes across redshift and cosmology using correlations with local halo environment.", *MNRAS*, doi:10.1093/mnras/stac2605

Overview of the talk

1. Background and motivation
2. Organisation of the thesis
3. Simulations and analysis techniques
4. Findings
5. Summary and future plans

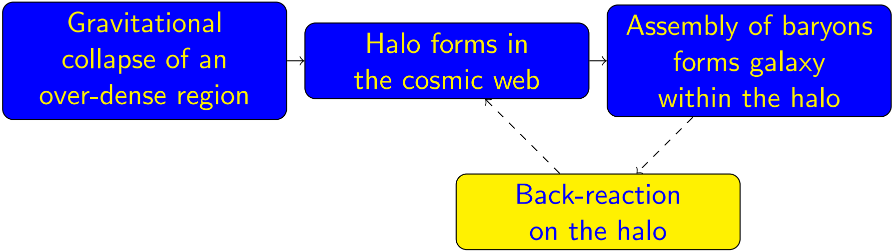
Introduction

- As matter clumps over overdensities due to gravity, the baryons cool and condense to form galaxies, leaving extended objects of dark matter haloes in the common paradigm of cosmology.
- These haloes are crucial in studying cosmology and dark matter particle physics, and are often modelled without any baryonic astrophysics using techniques like cosmological N-body simulations.
- Modeling galaxy formation does depend upon these haloes using for example semi-analytical techniques.

Gravitational collapse of an over-dense region → Halo forms in the cosmic web → Assembly of baryons forms galaxy within the halo

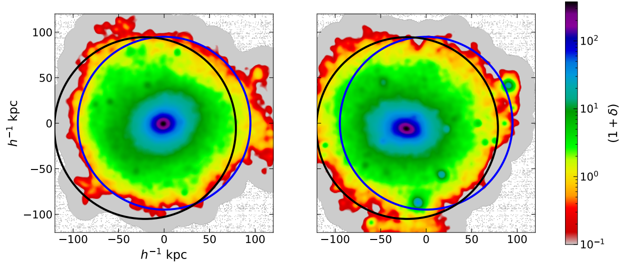
Introduction: Hydrodynamical simulations with galaxies

- Realistic galaxies are produced by modern hydrodynamical simulations of cosmological volumes with various subgrid baryonic prescriptions for the unresolved baryonic astrophysics such as in EAGLE, IllustrisTNG, Horizon, SIMBA, etc.
- In these simulations, the phase-space distribution of dark matter within the haloes have also been found to be significantly different and diverse indicating strong response to galaxies they host.



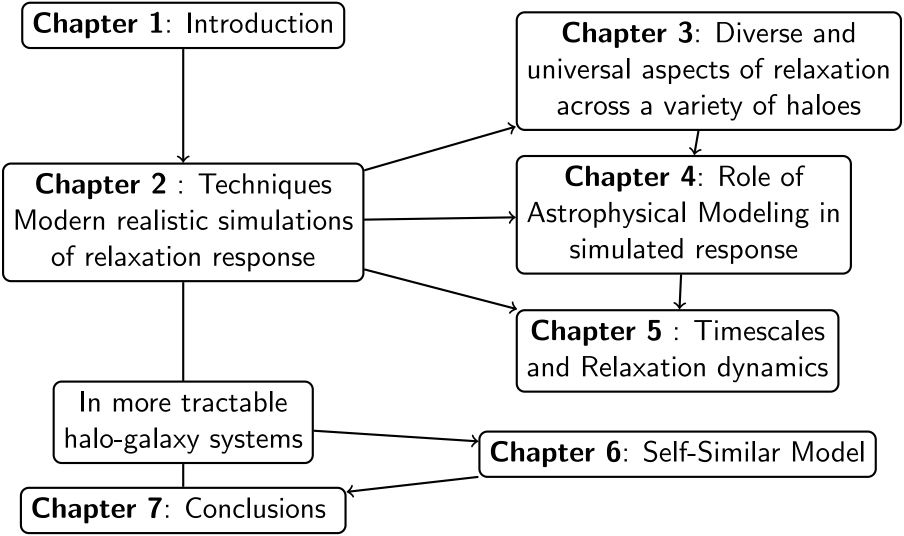
Dark matter halo response to galaxies

A halo from EAGLE simulation in the presence of galaxy (left image) can be seen more compact, spherical and even their centres shifted.



This thesis aims to understand the change in the radial mass distribution of dark matter in the haloes that influence key observables such as the rotation curves and radial acceleration relations.

Overview of thesis structure



Cosmological simulations with realistic galaxy formation

In this work the following publicly available cosmological simulations with state-of-the-art subgrid models for baryonic astrophysics were used:

Simulation Suite	Simulations	Box Size	Resolution
IllustrisTNG	TNG50	35 h^{-1}Mpc	2×2160^3
	TNG100	75 h^{-1}Mpc	2×1820^3
	TNG300	200 h^{-1}Mpc	2×2500^3
EAGLE (Reference)	L0100N1504	100 Mpc	2×1504^3
	L025N0752	25 Mpc	2×752^3
EAGLE	L025N0376_Ref	25 Mpc	2×376^3
(Physics Variations)	$+8$ variations	25 Mpc	2×376^3
	RecalHR	25	2×752^3
CAMELS	TNG 1P $(1 + 4 \times 10)$	25 h^{-1}Mpc	2×256^3

Halo Identification

- In IllustrisTNG and EAGLE simulations, dark matter haloes are identified using a 3D Friends-of-Friends (FoF) algorithm.
- Subhaloes within these FoF groups are identified using the SUBFIND algorithm, which detects gravitationally bound substructures.
- The central subhalo, located at the gravitational potential minimum, defines the halo center.
- Halo properties are characterized using the virial radius R_{200c} and mass M_{200c}.
- In CAMELS, haloes are found using ROCKSTAR, a 6D phase-space FoF algorithm, but characterized with the same mass and radius definitions.

Matching haloes to study dark matter response

- To study halo evolution in response to galaxy formation, haloes from the full hydrodynamical simulations are matched against the haloes found and gravity-only simulations of the same cosmological box.
- We trace the dark matter particles in a hydrodynamical halo to the initial volume and then look for the corresponding halo in the gravity-only run.
- To speed up the the match search, a KD-tree-based nearest neighbor search is used to pre-select candidate matches by ranking candidates based on spatial proximity and mass similarity.
- The matching approach achieves high completeness: over 96% of hydrodynamical haloes in IllustrisTNG and EAGLE simulations find a match in gravity-only runs.

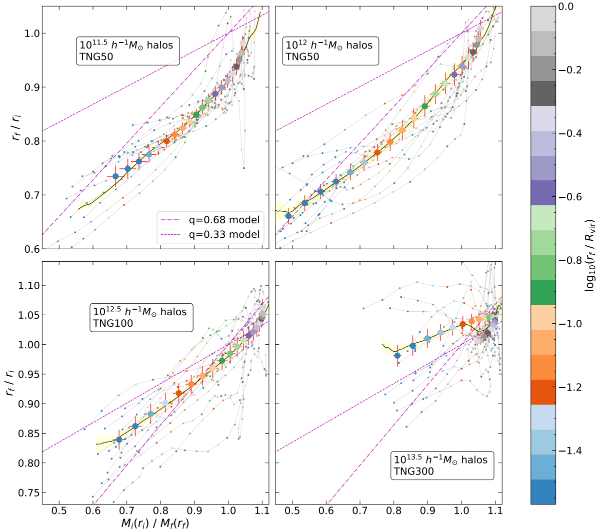
Physically characterizing the relaxation response

- Early works modeled this as adiabatic relaxation of dark matter in response to the **net** change in the gravitational potential due to galaxy formation.
- Consider a dark matter particle in circular orbit starting from radius r_i and adiabatically (conserving angular momentum) evolving to radius r_f caused by change in sphericalised total mass profile from $M_i(r)$ to $M_f(r)$ due to galaxy, then

$$r_i\, M_i(r_i) = r_f\, M_f(r_f) \implies \frac{r_f}{r_i} = \frac{M_i(r_i)}{M_f(r_f)}\,.$$

- By assuming no shell crossing, these quantities can be calculated by the comparing total and dark matter mass profile in the presence and absence of galaxies using the relation $M_f^d(r_f) = M_i^d(r_i)$.
- Quasi-adiabatic relaxation models consider the relaxation ratio r_f/r_i as a function of the mass ratio M_i/M_f.

Relaxation relation in IllustrisTNG

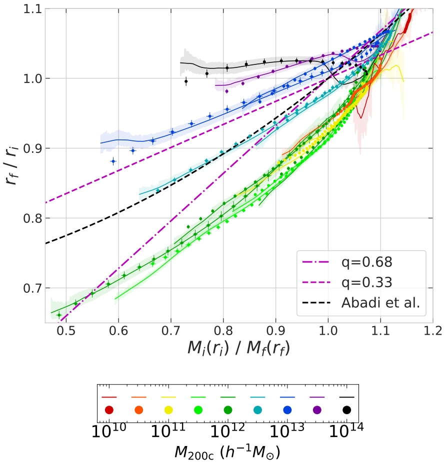


- Stacking the relaxation relation across haloes in populations selected by their sizes.
- Relaxation significantly weaker in cluster scale haloes.

Diversity in relaxation relation across halo masses

- Stacked relaxation relation is significantly different across halo masses.
- Existing quasi-adiabatic framework failed to provide a simple description across all scales.
- Relaxation significantly weaker in cluster scale haloes.

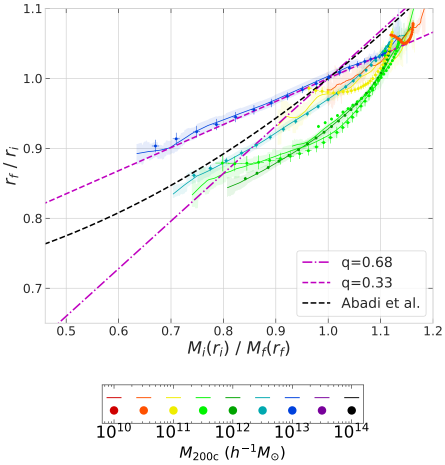
IllustrisTNG



Diversity in relaxation relation across halo masses

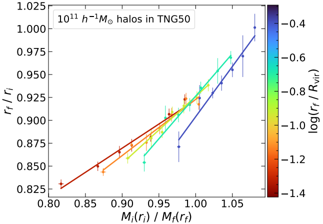
EAGLE

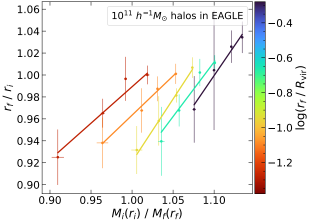
- Stacked relaxation relation is significantly different across halo masses.
- Existing quasi-adiabatic framework failed to provide a simple description across all scales.
- Relaxation significantly weaker in cluster scale haloes.

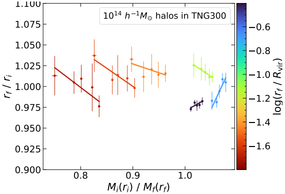


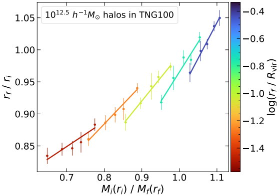
Dependence on halo centric distance

- A simple linear relation can accurately describe the relaxation, provided we assume an additional explicit dependence on the halo-centric distance.
- Notice that the exact nature of the relation is very different at different halo-centric distances (indicated by colorbar) for some halo masses, but they are all strongly linear.







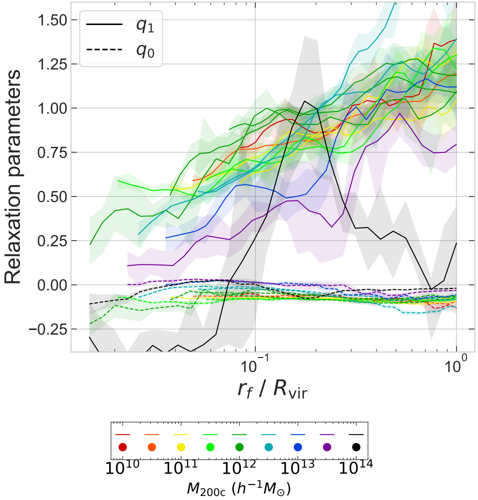


$$\frac{r_f}{r_i} - 1 = q_1(r_f)\left(\frac{M_i(r_i)}{M_f(r_f)} - 1\right) + q_0(r_f)$$

- Plotting the slope q_1 and y-intercept q_0 of the relaxation relation as a function of the halo-centric distance r_f/R_{vir}.
- Except for large clusters, the $q_1(r_f)$ is simply linearly increasing with $\log(r_f)$.

$$q_1(r_f) = q_{10} + q_{11}\log\left(r_f/R_{\rm vir}\right)$$

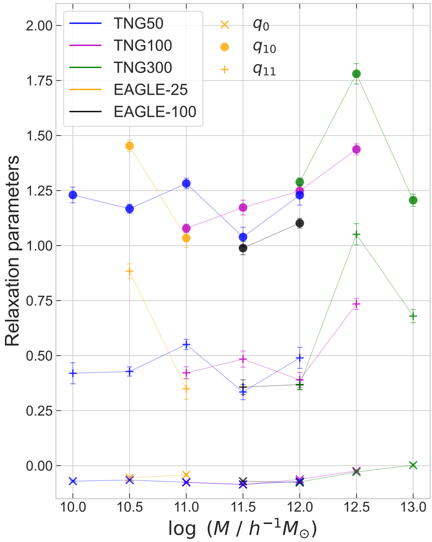
- This relation $q_1(r_f)$ is more universal than the radially independent relaxation relation.



Universal description of relaxation response

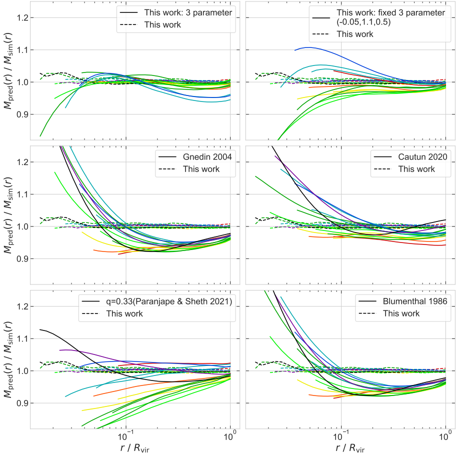
$$\frac{r_f}{r_i} - 1 = \left[q_{10} + q_{11} \log \left(\frac{r_f}{R_{\rm vir}} \right) \right] \left(\frac{M_i(r_i)}{M_f(r_f)} - 1 \right) + q_0$$

Three-parameter model (q_0, q_{10}, q_{11}) of relaxation is more universal across a wide range of halo masses.



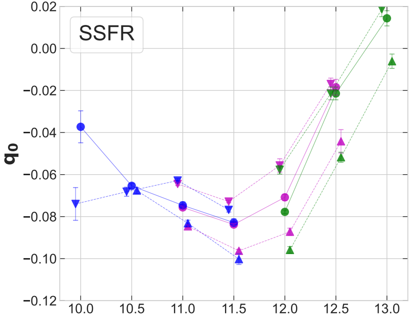
Improved predictions of relaxed dark matter profile

- Comparison of relaxed dark matter profiles from various models to IllustrisTNG simulation.
- Accounting for radial dependence of relaxation relation significantly improves the predictions shown by dashed curves in all panels.
- Even our 3-parameter model (q_0, q_{10}, q_{11}) (solid curves in top panels) is noticebly better than some of the existing models.



Astrophysical connection

- Focusing on the intercept offset q_0, which describes the amount of relaxation $r_f/r_i - 1$ for dark matter shells with no net change in the total enclosed mass $M_i/M_f = 1$.
- The value of q_0 today was found to be more negative among haloes hosting galaxies with higher specific star formation rate (SSFR).



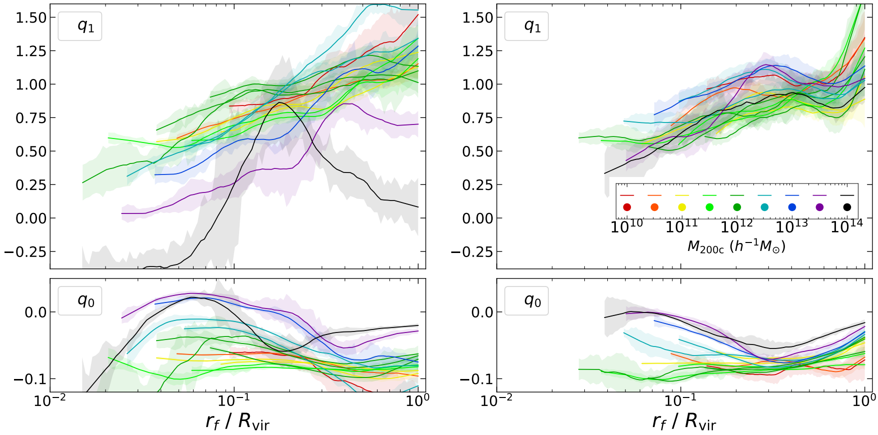
This higher excess relaxation might be related to larger amount of recent feedback output.

Chapter 3: Summary

- The relaxation response of dark matter haloes to galaxy formation is significantly diverse across halo masses when looking though the quasi-adiabatic framework.
- A simple linear relation can accurately describe the relaxation, provided we assume an additional explicit dependence on the halo-centric distance.
- This novel method of characterizing the radially dependent relaxation response is more universal than the standard radially independent relaxation relation.
- The relaxation offset q_0 was found to be more negative among haloes hosting galaxies with higher specific star formation rate.

Universal description of Halo Relaxation Response at $z = 1$

The radially dependent slope $q_1(r_f)$ and intercept $q_0(r_f)$ of this linear fit, is more universal across a wide range of halo masses up to $10^{13}h^{-1}M_\odot$ at $z = 0$ (left) and up to $10^{14}h^{-1}M_\odot$ at earlier redshift, $z = 1$ (right).



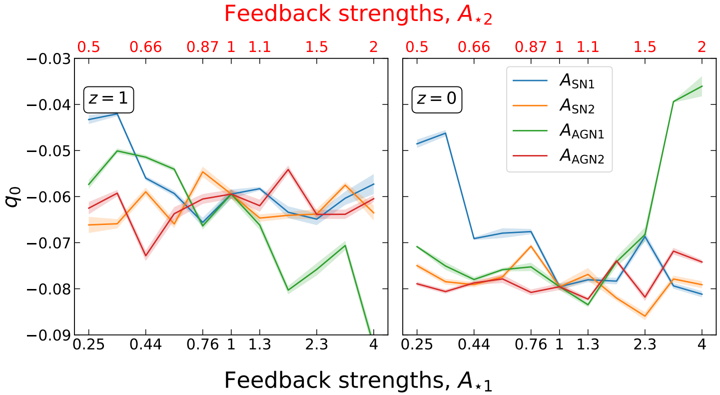
Astrophysical Feedback Variations in CAMELS

- From CAMELS (TNG 1P) we study the 41 hydrodynamical simulations.
- Uses IllustrisTNG model with varying baryonic feedback strengths.
- Two supernova feedback parameters: A_{SN1} and A_{SN2}.
- Two AGN feedback parameters: A_{AGN1} and A_{AGN2}.
- Simulations are limited in volume and resolution, affecting halo sample size.

Parameter	**Description**
A_{SN1}	Supernova energy injection, varied 0.25 to 4
A_{SN2}	Supernova wind speed, varied 0.5 to 2
A_{AGN1}	AGN feedback power, varied 0.25 to 4
A_{AGN2}	AGN wind speed & burstiness, varied 0.5 to 2

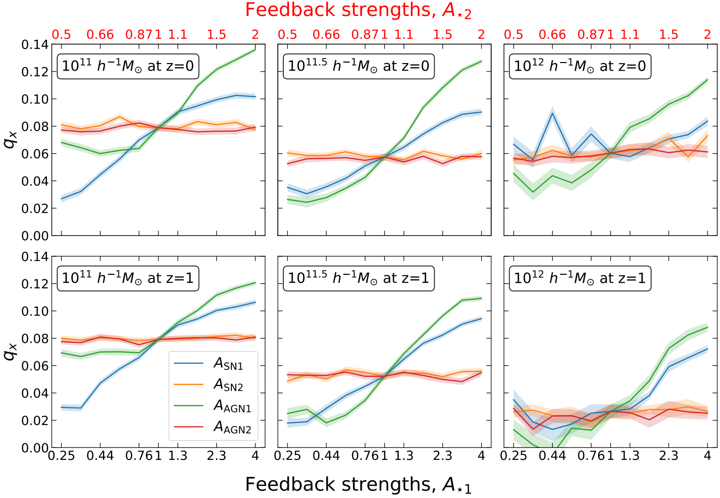
Relaxation Response in CAMELS

- Due to smaller box size consider all haloes in a wider mass range, $10^{11} - 10^{12} h^{-1} M_{\odot}$.
- Offset parameter q_0 examined due to resolution limits.
- Expected to be more negative with stronger feedback effects.



Relaxation Response in CAMELS: Mass dependence

- Intercept parameters: q_x: x-intercept measures mass ratio shift.
- Results suggest stronger feedback effects increase q_x, reducing q_0.



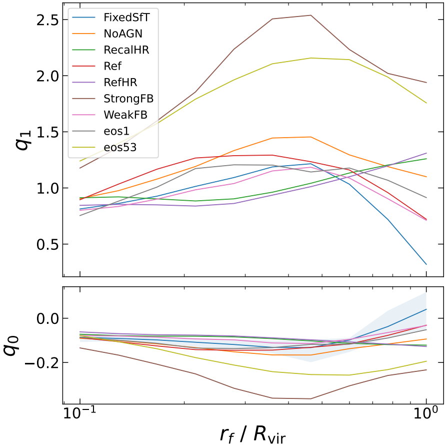
Astrophysical Model Variations in EAGLE

- EAGLE suite includes simulations with variation in baryonic subgrid physics such as the gas, star formation, and feedback models.

Simulation	Description
Ref	Standard EAGLE subgrid model in small box
eos53	Gas equation of state $\gamma = 5/3$
eos1	Isothermal equation of state $\gamma = 1$
FixedSfThresh	Constant star formation threshold
WeakFB	50% lower stellar feedback efficiency
StrongFB	$2\times$ higher stellar feedback efficiency
NoAGN	AGN feedback disabled
AGNdT8	AGN heating temperature $\Delta T_{\rm AGN} = 10^8$ K
AGNdT9	AGN heating temperature $\Delta T_{\rm AGN} = 10^9$ K
RecalHR	High-resolution variant with recalibrated parameters

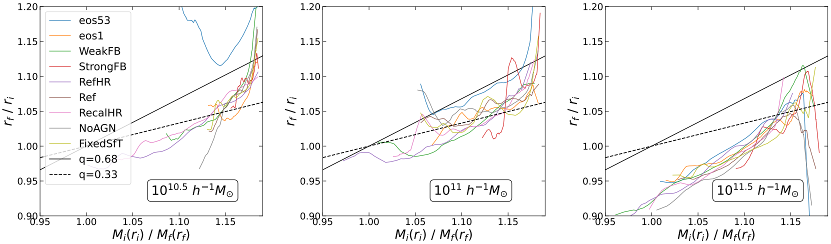
Relaxation Response in EAGLE

- Due to smaller box size consider all haloes in a wider mass range: $10^{10.5} - 10^{11.5} h^{-1} M_{\odot}$.
- Relaxation parameters analyzed as function of halo-centric distance.
- Strong feedback run shows the strongest deviation from the reference model.
- However higher gas equation state run also show similar level of deviation.



At different halo mass in EAGLE physics variation

- Alternatively consider three narrow mass bins around: $10^{10.5}h^{-1}M_\odot$, $10^{11}h^{-1}M_\odot$, and $10^{11.5}h^{-1}M_\odot$.
- The relaxation relation independent of halo-centric distance suggest that varying equation of state have strongest impact especially in the dwarf scale haloes.

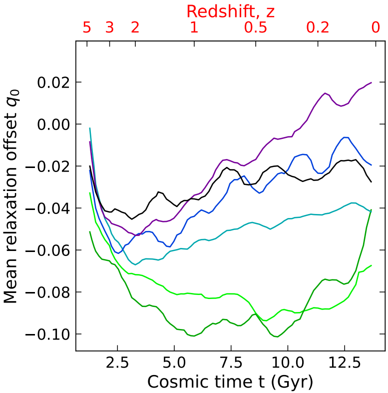


Chapter 4 Summary

- Relaxation response more universal at earlier epoch of $z = 1$ in IllustrisTNG
- In the CAMELS simulations with variation in feedback strengths, the relaxation offset strongly depends on the overall feedback flux from the galaxies. However, the parameters controlling the feedback burstiness and the wind speed have negligible impact on the relaxation offset.
- In the suite of EAGLE simulations with baryonic physics variation, the stronger feedback run does show a strong deviation in the relaxation from the reference model. However, the higher gas equation of state run show significantly strong impact on relaxation especially in the low mass haloes.

Relaxation Dynamics

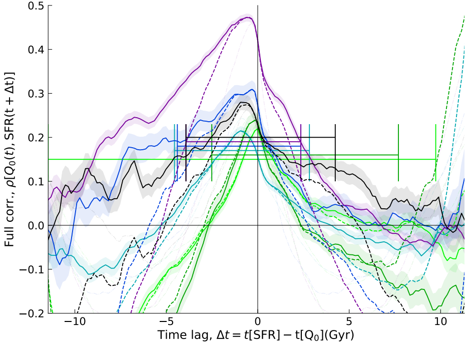
- Focusing on the intercept offset q_0, which describes the amount of relaxation $r_f/r_i - 1$ for dark matter shells with no net change in the total enclosed mass $M_i/M_f = 1$.
- It starts at zero, becomes more negative initially, but then apparently revert slowly back to zero.



Again a connection with star formation rate?

Temporal Connection with Astrophysics

- Spearman-rank correlation between $Q_0 \equiv -q_0$ and mean SFR across haloes over long periods of time is shown.
- This suggest that the offset is typically stronger following periods of high star formation activity.



Time Lag Analysis

- Spearman correlations ($\rho_h^s, \rho_t^s,\ \&\ \rho_f^s$) as a function of time lag (τ) depict a qualitative picture of the sequences of galactic processes.
- We compute the mean time lag between SFR and metallicity weighed by their correlation strengths.
- Spearman correlation coefficients in the correlation era ρ_+ and the anti-correlation era ρ_- are defined as:

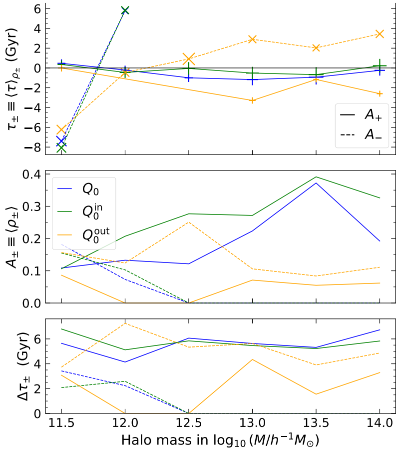
$$\rho_+(\tau) = \begin{cases} \rho_f^s(\tau) & \text{if } \rho_f^s(\tau) > 0 \\ 0 & \text{else} \end{cases}, \qquad \rho_-(\tau) = \begin{cases} -\rho_f^s(\tau) & \text{if } \rho_f^s(\tau) < 0 \\ 0 & \text{else} \end{cases}$$

$$\Delta\tau_+ = \int_{\rho(\tau)>0} w d\tau, \qquad \Delta\tau_- = \int_{\rho<0} w d\tau$$

$$A_+ = \int w(\tau)\rho_+(\tau)d\tau/\Delta\tau_+, \qquad A_- = \int w(\tau)\rho_-(\tau)d\tau/\Delta\tau_-$$

$$\tau_+ = \frac{1}{A_+\Delta\tau_+}\int \tau w(\tau)\rho_+(\tau)d\tau, \qquad \tau_- = \frac{1}{A_-\Delta\tau_-}\int \tau w(\tau)\rho_-(\tau)d\tau$$

Timescales of Relaxation Dynamics



- The correlation weighted mean time lags ($\tau_\pm$) between relaxation offset strengths ($Q_0, Q_0^{\rm in}, Q_0^{\rm out}$) against the star formation rate (SFR) vs final masses at $z \sim 0$ are shown.
- The corresponding strengths ($A_\pm$) and the durations ($\Delta\tau_\pm$) of the correlation are shown in the middle and bottom panels, respectively.
- The relaxation offset in the inner region is immediately affected by high SFR, while the outer regions are affected after around 2-3 Gyr.
- This likely explains the explicit dependence on the halo-centric distance.

Chapter 5 Summary

- The relaxation offset q_0 in the linear relaxation relation, describing the amount of relaxation $r_f/r_i - 1$ for dark matter shells with no net change in the total enclosed mass $M_i/M_f = 1$, is shown to be more negative following periods of high star formation activity.
- Using time correlation analysis, it is found that the offset in the inner region is immediately affected by high SFR, while the outer regions are affected after around 2-3 Gyr. This likely explains the explicit dependence on the halo-centric distance.

Self-Similar Model Introduction

- The most realistic description of halo relaxation response in the presence of baryons is provided by hydrodynamical simulations.
- In the previous chapters, we quantified this response in such simulations by studying the mass profiles of matched pairs of halos in the hydrodynamical and gravity-only runs of the simulation.
- In this chapter, we hunt for some more physical insights into the problem of halo relaxation due to baryonic backreaction using a more tractable, spherically symmetric self-similar model.
- The assumption of self-similarity reduces the nonlinear coupled partial differential equations into ordinary differential equations.
- We incorporate two novel additions: (i) a self-similar cooling of the gas and (ii) the formation of a pseudo-disk of gas along with the self-similar collapse of the dark matter halo.

Equation Setup: Self-Similar Evolution

- At any given time the radius of the shell undergoing turn around is used to set the scale of the system at that time.
- In terms of the self-similar radius $\lambda \equiv r/r_{\frown}(t)$, the spatial variation and the time evolution can be combined.
- The dimensionless self-similar profiles of velocity $V(\lambda)$, density $D(\lambda)$, mass $\mathcal{M}(\lambda)$ and pressure $P(\lambda)$ are then defined such that,

$$\rho = D(\lambda)\rho_H = \frac{D(\lambda)}{6\pi G t^2}, \qquad p = P(\lambda)\rho_H \left[\frac{r_{\frown}(t)}{t}\right]^2,$$

$$v = \frac{r_{\frown}(t)}{t} V(\lambda), \quad M = \frac{4\pi\rho_{\rm H}}{3}(r_{\frown}(t))^3\,\mathcal{M}(\lambda) \quad \text{with} \quad r_{\frown}(t) \propto t^{\delta};$$

Equation Setup: Self-Similar Evolution

When including both pressureless dark matter and gas, their evolution is coupled by gravity:

$$\frac{\mathrm{d}\ln}{\mathrm{d}\ln\lambda}\begin{bmatrix}-\bar{V}\\ D\\ P\end{bmatrix} = \frac{1}{\bar{V}^2}\frac{1}{\gamma\mathcal{T}-1}\begin{bmatrix}-\gamma\mathcal{T} & 1 & -\mathcal{T}\\ 1 & -1 & \mathcal{T}\\ \gamma & -\gamma & 1\end{bmatrix}$$
$$\times\begin{bmatrix}2\bar{V}(V-\lambda)\\ \frac{2}{9}(\mathcal{M}_g(\lambda)+\mathcal{M}_d(\lambda))\lambda^{-1}+(\delta-1)V\lambda-10V(\lambda/\lambda_{\mathrm{d}})^{-10}\\ \bar{V}\lambda[(2-\bar{\Lambda}_0 D^{(2-\nu)}P^{\nu-1})(\gamma-1)+2(\delta-1)]\end{bmatrix}$$
$$-\begin{bmatrix}\delta\lambda/\bar{V}\\ 0\\ 0\end{bmatrix}$$

Dark Matter Shell Evolution

Dark matter shell evolution is influenced by the total enclosed mass:

$$\frac{\mathrm{d}^2\lambda}{\mathrm{d}\xi^2} + (2\delta - 1)\frac{\mathrm{d}\lambda}{\mathrm{d}\xi} + \delta(\delta - 1)\lambda = -\frac{2}{9}\frac{(\mathcal{M}_g(\lambda) + \mathcal{M}_d(\lambda))}{\lambda^2} \tag{1}$$

- These two equation are solved iteratively to obtain the self-consistent evolution of gas and dark matter accreting self-similarly onto galaxy and halo respectively.

Parameters in the model:

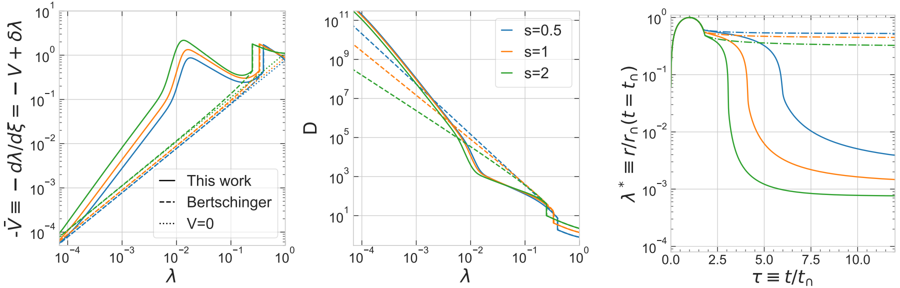
Parameters in the self-similar model of cooling and accretion of shocked gas onto a galaxy disk-like structure co-evolving with the self-similar collapse of host dark matter halo include the following:

Parameter	Description	reference value	variation
s	mass accretion rate	1	0.5 to 3
γ	gas equation of state	5/3	1.33 to 2
$\lambda_{\rm s}$	shock radius	0.9 $\lambda_{\rm sp}$	0.5 - 1.1 $\lambda_{\rm sp}$
f_b	baryon fraction within turnaround	0.1568	–
$\bar{\Lambda}_0$	dimensionless cooling rate	0.03	10^{-3} to 0.3
ν	cooling function slope parameter	1/2	
$\lambda_{\rm d}$	radius of disk-like model galaxy	5% $\lambda_{\rm s}$	2 - 25 % $\lambda_{\rm s}$

Table: Parameters in the self-similar model of cooling and accretion of shock gas onto a galaxy disk-like structure co-evolving with the self-similar collapse of host dark matter halo. Here $\lambda_{\rm sp}$ is the corresponding splashback radius of dark matter.

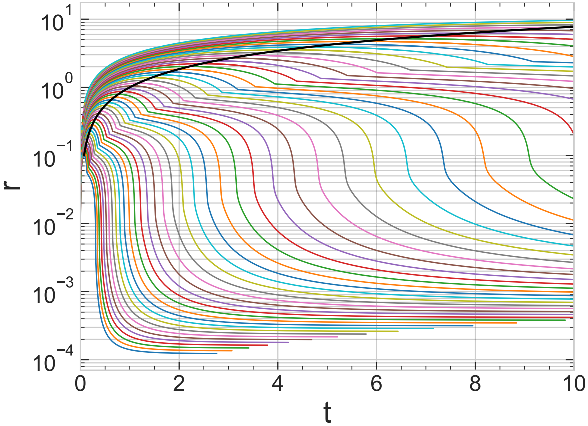
Self-similar accretion of gas

- Self-similar accretion of gas forming a galactic disk-like structure.



Self-similar accretion of gas

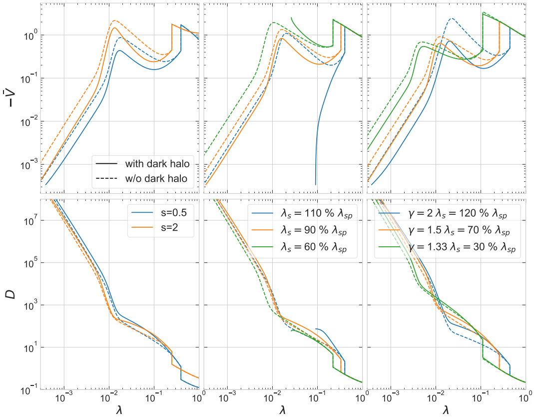
Self-similar accretion of gas forming a galactic disk-like structure that evolves self-similarly with the turn around radius of the host dark halo.



Effect of background halo on gas accretion

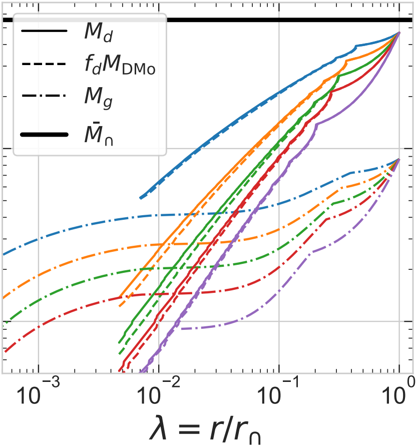
Velocity (top) and density (bottom) profiles of self-similar gas co-evolved with an accreting dark matter halo (solid lines) compared to profiles without halo (dashed lines).

- Left: Different accretion rates.
- Middle: Gas shocked at different radii.
- Right: Gas with different equations of state shocked at different radii.
- Host halo usually slows gas accretion, except when shock fails to slow down the gas strongly.

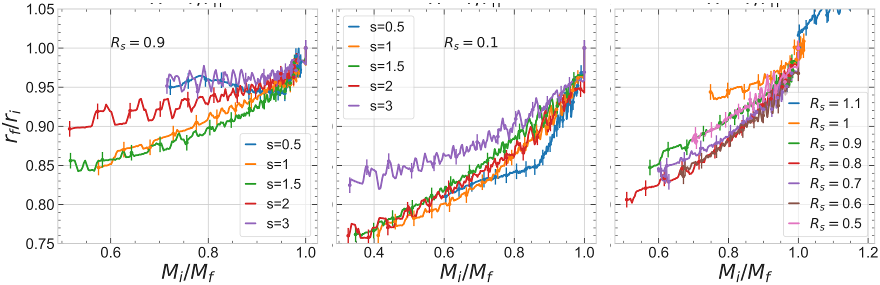


Effect of galaxy-like structure on the dark matter halo

- Effect of gas accreting onto a galaxy-like structure on the dark matter halo can be ascertained similarly by comparing the dark matter profiles with and without the galaxy.
- Compared to the dashed curves, the solid curves show that the dark matter halo is more relaxed by contracting towards the centre in the presence of the galaxy.

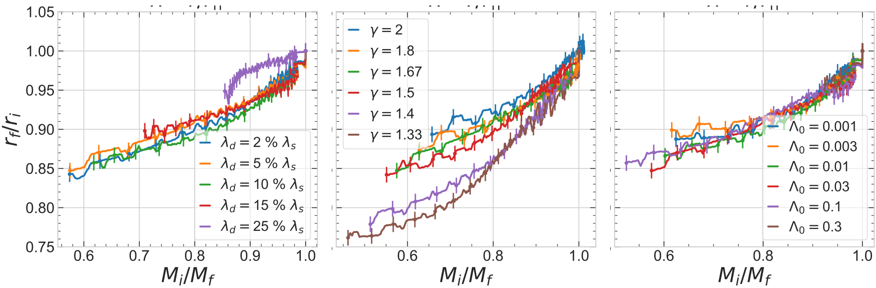


Effect of accretion mode and shock location on the relaxation



In the left, the accretion rate is varied while keeping the rest of the parameters fixed at reference values. In the middle, the accretion rate is varied specifically for the case of cold mode accretion. In the right, the shock location is varied.

Effect of gas and galaxy properties on relaxation



- In the left, the radius of the galaxy-like structure is varied, while the middle and right columns show variations in the gas equation of state and cooling rate, respectively.
- The gas equation of state has the strongest effect on the relaxation relation, with stiffer gas causing more relaxation as seen in the realistic simulations of EAGLE.

Chapter 6 Summary

- We propose a self-similar model for the relaxation response of dark matter haloes to galaxy formation.
- The model is used to study the effect of gas accretion onto a galaxy-like structure on the dark matter halo.
- The gas equation of state has the strongest effect on the relaxation relation, with stiffer gas causing more relaxation as seen in the realistic simulations of EAGLE.

Summary

- The relaxation response of dark matter haloes to galaxy formation is diverse across halo masses and can be accurately described by a linear relation with an explicit dependence on halo-centric distance.
- The relaxation offset q_0 is more negative following periods of high star formation activity, with immediate effects in the inner regions and delayed effects in the outer regions explaining the dependence on halo-centric distance.
- The relaxation response is more universal at earlier epochs and is strongly affected by the feedback flux from the galaxies.
- Additionally the gas equation of state has a significant impact on the relaxation response both in realistic simulations and also in more tractable self-similar systems.

Future plans

- Use direct probe of feedback to understand the role of AGN and other feedback on the relaxation dynamics.
- Use analytical tools and semi-analytical experiments to build an entirely physical and accurate model of relaxation.
- Run galaxy forming cosmological simulations specially focussed on studying the dark matter halo relaxation response.
- Identify galaxy properties that keeps record of evolutionary history of galactic processes allowing accurate determination of dark halo response.

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Thank You

For letting me share some of the correlations I made with the Universe, hope this increased your overall correlation with the elegant workings of the cosmos!